# Appendix K - Block Diagrams

Schematic-level layout of the three internal buses that connect the chips inside the Intuition Engine: the video compositor, the audio mixer, and the main system bus. Each diagram is drawn at the granularity a programmer needs to reason about cross-chip interactions, not at the granularity an engineer would use to fabricate the silicon.

## K.1 The video compositor

The compositor has six layers. Each layer comes from one of the six video engines. Layers are listed in back-to-front order; a higher layer covers the lower ones where it draws non-transparent pixels.

```
                  +-----------------------------+
                  |  HDMI / display output      |
                  +--------------^--------------+
                                 |
                  +--------------+--------------+
                  |   Video compositor          |
                  |   (per-pixel layer mux,     |
                  |    palette, blend modes)    |
                  +-+----+----+----+----+----+--+
                    |    |    |    |    |    |
         layer 20   |    |    |    |    |    |   (front)
                    |    |    |    |    |    +-- Voodoo 3D
                    |    |    |    |    +------- ULA
                    |    |    |    +----------- ANTIC + GTIA
                    |    |    +----------------- TED video
                    |    +--------------------- VGA
                    +-------------------------- VideoChip (layer 0, back)
```

Each engine writes into its own framebuffer (in main VRAM or in its private aperture); the compositor reads all six on every output line and resolves them into a single stream of RGBA pixels.

All six sources are collected on the same `60` Hz frame cadence. A chip that has scanline work still completes that work in its own frame before the compositor resolves the stack.

## K.2 The audio mixer

The audio mixer has separate engine inputs, plus the SFX channels, and one stereo output stream.

```
SoundChip --+
PSG / AY --+
SN76489 ---+
SID  ------+
SID2 ------+
SID3 ------+
POKEY -----+--->  Mixer  --->  Overdrive ---> Filter ---> Reverb ---> Output
TED audio -+      (sum,         (global)      (global)    (global)
AHX -------+       per-chip      voice)
MIDI/MUS --+       gain,
Live MIDI -+       per-chip
MOD -------+       mute)
WAV -------+
Paula DMA -+
           |
SFX ch 0-3 +
```

The SoundChip's own filter, the SID family's resonant filter, and the engine-internal effects all feed the mix before the global overdrive / filter / reverb stage; the global effects apply once per output sample to the summed signal.

MIDI/MUS and Live MIDI share the RawlandMini synth and voice pool. The diagram shows both control surfaces because one starts stored song data and the other streams immediate MIDI bytes.

## K.3 The system bus

Every CPU and every device hangs off one shared `64`-bit physical bus. The bus carries `64`-bit physical addresses and supports `8`, `16`, `32`, and `64`-bit transfers through CPU and device adapters. It arbitrates simultaneous accesses on a round-robin basis; there is no priority encoding above the CPU / DMA distinction.

```
                  +--------------------+
                  |    Main RAM        |
                  +---------^----------+
                            |
                            |
    +------+   +------+   +--+--+   +------+   +------+   +------+
    | IE64 |   | IE32 |   | M68K|   | 6502 |   | Z80  |   | x86  |
    +--+---+   +--+---+   +--+--+   +--+---+   +--+---+   +--+---+
       |          |          |         |          |          |
       +----------+----------+--+------+----------+----------+
                                |
                  +-----------+----------+
                  |   System bus          |
                  |   (64-bit address,    |
                  |    8/16/32/64-bit     |
                  |    transfers,         |
                  |    arbitrated)        |
                  +-+---+---+---+---+---+--+
                    |   |   |   |   |   |
              MMIO  |   |   |   |   |   |  DMA engines:
              regs  |   |   |   |   |   |   - blitter
                    |   |   |   |   |   |   - Amiga Paula DMA
              VGA --+   |   |   |   |   +-- - SFX sample DMA
              Video chip|   |   |   |       - copper
              ULA ------+   |   |   |
              Audio --------+   |   |
              File I/O ---------+   |
              Coprocessor ---------+
```

IE32, M68K, and x86 are 32-bit bus clients. The 8-bit CPUs (6502, Z80) reach the bus through an address translator that turns their 16-bit address space into selected low-window bus addresses, with the bank registers described in Chapters 27 and 28 selecting which translation applies.

## K.4 Coprocessor channels

The coprocessor block (Chapter 32) is a many-to-many channel between the main CPU and a pool of worker CPUs. Each worker listens on its own ring buffer; the main CPU posts work items into the ring and reads the result back.

```
    Main CPU
    |
    |  COSTART / COCALL  -->  +----------------+
    |                         |  Ring buffer   |
    |  COSTATUS / COWAIT  <-- |  (per-worker)  |
    |                         +-------+--------+
    |                                 |
    |                         +-------+--------+
    |                         |  Worker CPU    |
    |                         |  (one of 6     |
    |                         |   types)       |
    |                         +----------------+
```

Six worker types share the protocol: an IE64 worker, an IE32 worker, a 6502 worker, a Z80 worker, an M68K worker, and an x86 worker. The main CPU is whichever one boots the machine; all the others are available as workers when not booted.

## K.5 Bus translation for the small CPUs

The 6502 and Z80 see the same overall bus through a 16-bit-to- 64-bit translator. In practice their documented apertures target the low memory and MMIO window. The translator has two independent functions:

```
6502 / Z80 address (16-bit)
     |
     v
+----+-----------------------+
| Decode page                |
|  - bank registers $F7xx? --> intercepted, never reach bus
|  - MMIO mirror $F000-$FFF9? -> +$F0000 -> bus
|  - $E000-$EFFF banked window-> + bank * 4 KB -> bus
|  - everything else --------- > main RAM page
+----+-----------------------+
     |
     v
bus address (64-bit, low-window value here)
```

The bank registers at `$F700-$F705` (low/hi pairs for apertures 1, 2, 3) and `$F7F0` (aperture 0) are captured before the translator runs. A write to `$F700` does not reach `$F0700`; it loads the low byte of bank register 1.